Hugo Hadjur

PhD in Computer Science — Data Science & AI Expert

hugo.hadjur@saryga.com

Academic Page   Google Scholar   LinkedIn

I have a **PhD** in informatics with **ENS de Lyon** with strong dominance in **AI**, and **NLP** experience.
Since 2023, I am working in applied **biostatistics and AI research** for **pharma and biotech**.
I **write peer-reviewed scientific papers, find and adapt state-of-the-art methods and collaborate with researchers outside my field**.

Work Experience

Since Dec. 2023
Permanent

AI & Statistical Methodologist at **Saryga** (France)

— **AI & innovation projects projects**: design and implementation of LLM- and NLP-based solutions (semantic analysis, text summarization, zero-shot classification, paraphrase mining) to support data analysis workflows; development of interactive applications (R Shiny) and R packages for decision support.
— **Biostatistics**: apply and adapt state-of-the-art statistical and ML methods in clinical development (highly-regulated), lead and co-author peer-reviewed publications, and ensure traceability and transparency of methods and results.
— **Mentoring**: supervise interns and contribute to the internal AI community through code review, methodological guidance, and informal training sessions.

Sept. 2020 - Aug. 2023
3-year contract

PhD Candidate and Assistant Professor at **École normale supérieure de Lyon** and **aivancity** (France)

PhD Thesis: **Designing sustainable, autonomous, and energy efficient Internet of Things systems, applied to smart beehives**
Advisorship of Laurent Lefèvre & Doreid Ammar, at LIP-Avalon laboratory

— **Research & Innovation**: explored reinforcement learning and distributed AI to optimise energy-aware IoT services. Innovation approach: development of POC, simulations, prototype (MVP), and deployment in real beehives (test & learn).
— **Teaching**: designed and delivered courses in advanced data analysis and visualisation, data preprocessing and coding for AI and data science (Python), to mixed cohorts of students and professionals.
— **Project & Management**: supervised interns on connected beehives deployment and data analysis; headed the online course platform aivancityX (coordination with stakeholders, content updates, user support).

Sep. 2019 - Aug. 2020
& Oct. 2018 - Dec. 2018
1-year contract

Instructor & Researcher at **emlyon Business School** (France)

— **Research**: data analysis for connected beehives and bee behaviour (IoT, machine learning, computer vision).

— **Teaching & training**: led intensive Python programming bootcamps and the business analyst toolbelt (SQL, Excel, Tableau), training students to become autonomous in data analysis and visualisation.

Jan. 2019 - Sep. 2019
8-month contract

Data Analyst at **Schneider Electric**, Houston, TX (USA)

— Supported the business by building reports/dashboards and KPIs to monitor performance, conducted database integration
— Communicated findings and insights to cross-functional team members and management
— Tools: Python, SQL, Tableau

Oct. 2017 - Sep. 2018
1-year exchange

Student Researcher at **Kyoto University's Ishii Lab** (Japan)

Master's thesis: master board games thanks to Reinforcement Learning (Q-Learning, SARSA, Deep Reinforcement Learning)

May-Sep. 2017
4-month internship

Data Science Research Internship at **CiNet** (Center for Information and Neural Networks, Osaka, Japan)
— Conducted research about the links between Twitter data, human behavior and brain data, resulting in a published peer-reviewed paper
— Analyzed large data sets (≈ 500,000 Twitter users) using clustering & regression techniques.
— Worked in an international, multicultural environment (Japan/Europe).

Publications

[under revision] **Hugo Hadjur**, Julia Geronimi, Sandrine Guilleminot, Alessandra Serra, Libby Daniells, Marie-Karelle Riviere, Gaëlle Saint-Hilary, Pavel Mozgunov. **Evaluating methods for biomarker cut-off detection: a comparative study in small and unbalanced samples**, Statistics in Medicine, 2026

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Deep Reinforcement Learning for Energy-efficient Selection of Embedded Services at the Edge**, 2024 IEEE International Conferences on Internet of Things (iThings), Copenhagen, Denmark, 2024, pp. 67-74, 10.1109/iThings-GreenCom-CPSCom-SmartData-Cybermatics62450.2024.00034. hal-04708697

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Services Orchestration at the Edge and in the Cloud for Energy-Aware Precision Beekeeping Systems**, 2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), St. Petersburg, FL, USA, 2023, pp. 769-776, 10.1109/IPDPSW59300.2023.00129. hal-04091575

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Toward an intelligent and efficient beehive: A survey of precision beekeeping systems and services**, Computers and Electronics in Agriculture, Elsevier, 2022, 192, pp. 1-16. 10.1016/j.compag.2021.106604. hal-03483914v2

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Analysis of Energy Consumption in a Precision Beekeeping System**. IoT '20 - 10th International Conference on the Internet of Things, Oct 2020, Malmö, Sweden. pp. 1-11, 10.1145/3410992.3411010. hal-02973772

Kazuma Mori, **Hugo Hadjur**, Masahiko Haruno. **Natural Language Content Mediates the Association Between Active Interactions on Social Network Services and Subjective Well-Being**, Cyberpsychology, Behavior, and Social Networking, 2022, pp. 678-685. 10.1089/cyber.2021.0340

Alessandra Serra, Julia Geronimi, Sandrine Guilleminot, **Hugo Hadjur**, Marie-Karelle Riviere, Gaëlle Saint-Hilary, Pavel Mozgunov. **A Novel Approach to Assess the Predictiveness of a Continuous Biomarker in Early Phases of Drug Development**, Statistics in Medicine, 2025, 44: e70026. 10.1002/sim.70026

Communication

Posters

— **SnB** (Statistics and Biopharmacy Conference) - Paris, France, 2025. Evaluating methods for biomarker cutoff detection: a comparative study in small and unbalanced samples

— **ISCB** (Annual Conference of the International Society for Clinical Biostatistics) - Thessaloniki, Greece, 2024. Simple Approaches for Portfolio Quantitative Decision-Making

Presentations

— **PSI 2026** (Statisticians in the Pharmaceutical Industry Conference) - Belfast, Northern Ireland, 2026. Evaluating methods for biomarker cut-off detection

— **GreenDays 2023** - Lyon, France, 2023. Optimize task selection under energy constraints in IoT systems thanks to reinforcement learning

— **Journées du GDR Réseaux et Systèmes Distribués** - Rennes, France, 2022. Designing connected, sustainable, autonomous and low-consumption systems for a distributed network of smart beehives

— **IoT'20** - Online, 2020. Analysis of Energy Consumption in a Precision Beekeeping System

Technical skills

Programming:	**Python (PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, Natural Language Toolkit NLTK), R, SQL**
Tech Experience:	**Large Language Models (LLMs), Transformers, Natural Language Processing (NLP), Reinforcement Learning (RL), Computer Vision, Diffusion Models, Model Optimization, High-performance Computing (HPC)**
Design & Text:	**LaTeX, Microsoft Office, Inkscape, Adobe Photoshop**

Education

"Grande École" diploma + PhD

PhD in Informatics — ENS de Lyon, LIP-Avalon Laboratory - France, 2020-2023

Exchange year — Kyoto University Graduate School of Informatics - Japan, 2017-2018

Diplôme d'ingénieur — Ensimag, Grenoble INP - France, 2015-2018

Scientific preparatory classes — CPP Grenoble - France, 2013-2015

Scientific baccalaureate — French schools in Tokyo and Singapore - Japan / Singapore, 2007-2013

Languages

French: native

English: professional working proficiency

Japanese: intermediate